

Prithvi Harish Malhotra

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building AI developer tools, backend systems, distributed platforms, and cloud infrastructure. Skilled in Python, Java, C++, Model Context Protocol (MCP), Amazon Web Services (AWS), Google Cloud Platform (GCP), Kubernetes, and Terraform, with experience improving developer productivity, observability, and release workflows. Built enterprise AI developer tooling that improved **engineer productivity by 30%** and avoided **~\$100K/year in licensing costs**.

TECHNICAL SKILLS

Languages: Python, Java, C++, GO, C#, Bash, SQL

AI & Developer Tools: MCP, LangChain, Google Vertex AI

Cloud: AWS — ECS, EKS, CodeDeploy, S3, Athena, Lambda, CloudWatch | GCP — Cloud Run

DevOps & Infrastructure: Git, Docker, Kubernetes, Terraform, GitHub Actions, Concourse, Jenkins, ArgoCD, Ansible, Continuous Integration/Continuous Delivery (CI/CD)

Backend & Data: Spring Boot, REST APIs, PostgreSQL, Elasticsearch, Apache Kafka

Systems & Observability: Linux, KVM, Selenium, KEDA, Istio, Prometheus, Observe

PROFESSIONAL EXPERIENCE

Software Engineer, Rokt US Corp.

July 2025 – Aug 2026 | Seattle, USA

AI & Developer Tooling

- **Architected and led end-to-end development of an in-house Observe MCP**, enabling natural-language observability queries with privilege-aware dataset access; **improved engineer productivity by 30%** and **avoided ~\$100K/year in licensing costs**.
- **Architected and productized Rokt AI Code Guru**, an MCP-based developer tool using Python, LangChain, and Google Vertex AI to automate codebase context retrieval, handling **500–700 requests/day**.
- Engineered secure MCP integrations by unifying transport layers across distributed services, implementing custom JSON Web Token (JWT) authentication, and enforcing zero-trust access control.

Build & Release/CI/CD

- **Reduced CI/CD time from ~25 to 6–7 minutes pre-merge and ~20 to ~5 minutes post-merge** through selective C# builds; deployed services to **AWS ECS via CodeDeploy blue/green deployments**.
- **Architected autoscaling for GitHub Actions and CI runners** using KEDA and Karpenter, cutting PR-check wait time from **3–4 minutes to <15 seconds**.
- **Supported the migration from AWS ECS/CodeDeploy to EKS**, implementing **ArgoCD-based GitOps delivery** for Kubernetes services.

Research Assistant, GHOST Research Group

Sept 2024 – June 2025 | Boulder, USA

- **Designed and Led the end-to-end** development of a distributed orchestrator-worker system using Java and Spring Boot, coordinating 8 Selenium-driven KVM virtual machines to emulate high-fidelity, concurrent population web behavior for critical military network simulation research.
- Built a high-performance AWS serverless analytics pipeline (S3, Athena, Lambda, CloudWatch) for WiFi telemetry data, reducing end-to-end processing times by 95%.
- Automated AWS infrastructure and deployments using **Terraform, Jenkins, Docker, and Ansible**, reducing manual deployment and configuration time by **70%**.

Software Development Engineer, LTIMindtree

July 2021 - June 2023 | Pune, India

- Built transaction-processing systems for **HSBC**, transforming and validating incoming **XML and SIF transactions** into HSBC's standardized internal format for downstream processing.
- Developed **20+ enterprise-grade Java/Spring Boot REST APIs** to support mission-critical financial workflows; optimized complex PostgreSQL queries, schemas, and indexing strategies to boost **database performance by 30%**.
- **Led a cross-functional engineering team of 4** to establish automated testing frameworks using JUnit, Mockito, and Cucumber, **driving a 40% reduction** in post-deployment production defects.

EDUCATION

Master of Science in Computer Science, University of Colorado Boulder

May 2025 | USA

Bachelor of Technology in Electronics, Mumbai University

May 2021 | India

CERTIFICATIONS

- AWS Certified Developer - Associate
- Full Stack Training Certificate by Larsen and Toubro Infotech (Oct 2021)